Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Second Term
Week	9
Topic	Simplification of Algebraic Terms
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems
Theme	—
Curriculum objectives	• carry out order of operation in algebra • insert/remove bracket and simplify expression • solve word problems involving algebraic expression
Previous knowledge	• Addition and subtraction of algebraic terms • Multiplication and division of algebraic terms
Lesson sequence	• carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$ • insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$ • solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.
Instructional materials	• Chart showing algebraic expression.
Evaluation	• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ • Simplify $7 \times 3a - (3a + 5a) \times 2$

	• Simplify $12-(7-2f)-(3-f)$ • Simplify $(8a-5)+(14-3a)$ • A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Lesson objectives

1.	carry out order of operation in algebra
2.	insert/remove bracket and simplify expression
3.	solve word problems involving algebraic expression

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Simplification of algebraic expressions: order of operations, brackets and word problems

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: Getting the Order Right in Algebra

When an algebraic expression mixes several operations, you cannot work from left to right in any order you like. The result depends on the order in which you carry them out. So algebra follows one agreed set of rules.

Do the operations in this order:

1. Brackets first — work out what is inside them.
2. Order / Of — powers like ² or ³.
3. Division and Multiplication — from left to right.
4. Addition and Subtraction — from left to right.

A handy way to remember the order is BODMAS: Brackets, Of/Order, Division, Multiplication, Addition, Subtraction.

The trick in algebra is that you first deal with multiplication and division wherever they appear, before you do any adding or subtracting. Then you combine like terms. Let's see this in action.

Worked example 1

Simplify 4n + 3n × 10.

Here there are no brackets and no powers, so Division and Multiplication come first. The only multiplication is 3n × 10.

3n × 10 = 30n

So 4n + 3n × 10 = 4n + 30n.

Now do the addition: 4n + 30n = 34n.

Answer: 34n

Notice that if you had added 4n + 3n first (getting 7n) and then multiplied by 10, you would have got 70n — which is wrong. The multiplication must be done before the addition.

Worked example 2

Simplify 32 + 8x ÷ 2.

Again, there are no brackets. Division and Multiplication come before Addition and Subtraction. The only division is 8x ÷ 2.

8x ÷ 2 = 4x

So 32 + 8x ÷ 2 = 32 + 4x.

There is nothing more to add — 32 and 4x are not like terms, so the answer is left as 32 + 4x.

Answer: 32 + 4x

Worked example 3

Simplify 3 × 5x + 4x ÷ 2.

No brackets, no powers. Division and Multiplication come first, from left to right.

3 × 5x = 15x, and 4x ÷ 2 = 2x.

Now do the addition: 15x + 2x = 17x.

Answer: 17x

Worked example 4

Simplify 7 × 3a − (3a + 5a) × 2.

Brackets come first. Inside the bracket: 3a + 5a = 8a.

So the expression becomes 7 × 3a − 8a × 2.

Now Division and Multiplication come before the subtraction. Work from left to right:

7 × 3a = 21a, and 8a × 2 = 16a.

Now do the subtraction: 21a − 16a = 5a.

Answer: 5a

Now try these yourself. Work them out on paper first, then check your answers below. Remember to deal with brackets first, then multiplication and division, and only then addition and subtraction — combining like terms as you go.

Practice

1. Simplify 36y ÷ 6 + y + 2 × 4y.
2. Simplify 4x + 2 × 6x.
3. Simplify 5 × 3a − 2 × 3a.
4. Simplify (2y + 4y) × 3 − 14y.

Section 2: Removing and Adding Brackets

Brackets group terms together so that we know which parts of an expression belong together. When you want to simplify an expression that has brackets in it, the first thing to do is remove the brackets — and the sign just in front of a bracket decides how the terms inside come out.

Think of a bracket as a box. If a plus sign sits in front of the box, then every term inside the box keeps its own sign exactly as it is. You can lift the lid and write the terms down unchanged.

If a minus sign sits in front of the box, then the box is being subtracted from everything before it. To take the whole box away, every term inside must change sign: a plus becomes a minus, and a minus becomes a plus.

Once every bracket has been removed, you finish by collecting the like terms — the terms that carry the same letter.

Example 1: Simplify $(12a - 7) + (4 - 5a)$

There is a plus sign before the first bracket, so remove it unchanged: $12a - 7$.
There is also a plus sign before the second bracket, so remove it unchanged too: $4 - 5a$.

Put it all together:
$$12a - 7 + 4 - 5a$$

Now collect the like terms. The a-terms are $12a$ and $-5a$:
$$12a - 5a = 7a$$
The number terms are -7 and $+4$:
$$-7 + 4 = -3$$

So the answer is:
$$7a - 3$$

Example 2: Simplify $(8x - 2y) - (9x - 4y)$

The first bracket has a plus sign in front, so it comes out unchanged: $8x - 2y$.
The second bracket has a minus sign in front. Every term inside must change sign:
- $9x$ becomes $-9x$
- $-4y$ becomes $+4y$

So we write:
$$8x - 2y - 9x + 4y$$

Collect the like terms. The x-terms are $8x$ and $-9x$:
$$8x - 9x = -x$$
The y-terms are $-2y$ and $+4y$:
$$-2y + 4y = 2y$$

So the answer is:
$$-x + 2y$$

Practice

1. Simplify $12 - (7 - 2f) - (3 - f)$.
2. Simplify $(8a - 5) + (14 - 3a)$.
3. Simplify $(6p + 9) - (4p - 2)$.

Section 3: Turning Word Problems into Algebra

Word problems describe a situation in everyday words, and your job is to turn those words into an algebraic expression before you simplify it. To do that, look for two things: the unknown amount (the letter) and the relationship that connects the amounts.

Suppose a pen costs h naira and a crayon costs ₦20 less than a pen. The pen's cost is h, so the crayon costs h − 20. If the question asks for the cost of a crayon and a pen together, you add the two: h + (h − 20). Now simplify by collecting like terms: h + h − 20 = 2h − 20. So both items cost (2h − 20) naira.

The key steps are always the same:
1. Decide what the letter stands for (the unknown amount).
2. Write the other amount in terms of that letter, using the words that tell you the relationship — "less than" means subtract, "more than" means add, "twice" means multiply by 2.
3. Combine the expressions you have written and simplify by collecting like terms.

Envelopes cost d naira per pack. Four notepads cost ₦20 more than a pack of envelopes. Find the total cost of a pack of envelopes and a notepad.

Step 1 — Decide what the letter stands for. The pack of envelopes costs d naira.

Step 2 — Write the other amount. The notepads cost ₦20 more than a pack of envelopes, so they cost d + 20 naira. Since there are four notepads and they cost the same each, one notepad costs (d + 20) ÷ 4.

Wait — read the question carefully. It asks for the total cost of a pack of envelopes and a notepad, so you need one notepad, not the set of four. One notepad costs (d + 20) ÷ 4, which we write as (d + 20)/4.

Step 3 — Add and simplify. Total = d + (d + 20)/4.

To add these you need a common denominator of 4. Write d as 4d/4:
Total = 4d/4 + (d + 20)/4 = (4d + d + 20)/4 = (5d + 20)/4.

So the total cost of a pack of envelopes and one notepad is (5d + 20)/4 naira.

Practice

1. A pen costs h naira and a crayon costs ₦20 less than a pen. (a) Find the cost of a crayon. (b) Find the total cost of a crayon and a pen.
2. A pen and a pencil altogether cost x naira. If a pen costs y naira, find the cost of 3 pencils.
3. Tunde has m marbles. His sister has twice as many marbles as Tunde. Write an expression for the total number of marbles they have together.

Check your work

Attempt every question before you read this page.

Getting the Order Right in Algebra

1. 15y

   No brackets. Division and multiplication come first:
   36y ÷ 6 = 6y, and 2 × 4y = 8y.
   The expression becomes 6y + y + 8y. All three are like terms, so add them: 6y + y + 8y = 15y.
2. 16x

Multiplication first: 2 × 6x = 12x.
Then 4x + 12x = 16x.

3. 9a

   Multiply first: 5 × 3a = 15a, and 2 × 3a = 6a.
   Then subtract: 15a − 6a = 9a.

4. 4y

   Brackets first: 2y + 4y = 6y.
   Then multiply: 6y × 3 = 18y.
   Then subtract: 18y − 14y = 4y.

Removing and Adding Brackets

1. $2 + 3f$

   The first term, 12, has no bracket. The bracket $(7 - 2f)$ has a minus sign in front, so change every sign: 7 becomes -7 and $-2f$ becomes $+2f$. The bracket $(3 - f)$ also has a minus sign in front, so 3 becomes -3 and $-f$ becomes $+f$.

   So we get:
   $$12 - 7 + 2f - 3 + f$$

   Collect the number terms: $12 - 7 - 3 = 2$.
   Collect the f-terms: $2f + f = 3f$.

   Answer: $2 + 3f$.

2. $5a + 9$

   Both brackets have a plus sign in front, so remove them unchanged:
   $$8a - 5 + 14 - 3a$$

   Collect the a-terms: $8a - 3a = 5a$.
   Collect the number terms: $-5 + 14 = 9$.

   Answer: $5a + 9$.

3. $2p + 11$

   The first bracket has a plus sign in front, so it comes out unchanged: $6p + 9$.
   The second bracket has a minus sign in front, so change every sign: $4p$ becomes $-4p$ and -2 becomes $+2$.

   So we get:
   $$6p + 9 - 4p + 2$$

   Collect the p-terms: $6p - 4p = 2p$.
   Collect the number terms: $9 + 2 = 11$.

   Answer: $2p + 11$.

Turning Word Problems into Algebra

1. (a) h − 20 naira. (b) (2h − 20) naira.

   The pen costs h. The crayon costs ₦20 less, so it costs h − 20. Together they cost h + (h − 20) = h + h − 20 = 2h − 20 naira.

2. 3(x − y) naira.

   The pen and pencil together cost x, and the pen costs y. So one pencil costs x − y naira. Three pencils cost 3 × (x − y) = 3(x − y) naira.

3. 3m marbles.

   Tunde has m. His sister has twice as many, so she has 2m. Together they have m + 2m = 3m marbles.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-ef9fc79b	**Date**	
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____